Rahul Kumar Singh

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About Me

AI & Data Science student with hands on experience in developing and deploying real world AI applications. Passionate about solving practical problems through intelligent systems, with experience across Machine Learning, Computer Vision, and Generative AI. Strong foundation in programming, problem solving, databases, and software engineering, with a willingness to learn and work across emerging technologies and domains.

Skills

Languages: Python, C++, JavaScript,SQL

Programming & Software Engineering: OOPs, REST APIs, FastAPI, Flask, Node.js, Express.js, Websockets

Machine Learning: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Deep Learning & Computer Vision: TensorFlow, PyTorch, Keras, OpenCV, YOLOv8/YOLOv11, FaceNet, CNN

Generative AI: LLMs, LangChain, LangGraph, RAG, ChromaDB, Pinecone, OpenAI API, Gemini, Hugging Face

Databases: MongoDB, Firebase

Deployment & Tools: Git, GitHub, Docker, Vercel, Render

Experience

Machine Learning Intern Feb 2026 – May 2026

RVNS Solutions Pvt. Ltd. *Remote*

- Integrated a custom trained YOLOv8 model into an AI powered oral health analysis application for automated classification of five oral conditions.
- Developed backend integration across React.js, Node.js, Express.js, and Flask, connecting model inference with authentication and prediction history functionality.
- Supported cloud deployment across Vercel, Render, and Hugging Face Spaces to make the application accessible for user testing and inference.

Projects

AI Based Oral Health Analysis System Live Demo GitHub

Full Stack AI Healthcare, YOLOv8 2026

- Trained and integrated a custom YOLOv8 model to detect five oral conditions: cancer, calculus, caries, gingivitis, and ulcers.
- Implemented a three tier inference pipeline with confidence score tracking, MongoDB Atlas storage, and prediction history management.

AgentVerse – Multi Agent AI Chat Platform GitHub

LangGraph, FastAPI, Next.js, RAG 2026

- Developed a multi agent platform with specialized math, code generation, and general purpose agents, custom tool invocation, and persistent chat history.
- Built a RAG pipeline using ChromaDB and BGE-M3 embeddings for knowledge base question answering, with streamed responses via Server-Sent Events (SSE).

Drone Surveillance System GitHub

YOLOv11, FaceNet, WebSockets, PyBullet RL 2025

- Integrated object detection, facial recognition, and crowd density analysis across drone and CCTV video feeds.
- Implemented asynchronous communication with FastAPI and WebSockets and explored PPObased reinforcement learning for autonomous drone navigation in a PyBullet simulation.

Education

Dr. D. Y. Patil Institute of Engineering, Management & Research (DYPIEMR) 2023 – 2027

B.E. in Artificial Intelligence & Data Science, Pune CGPA: 8.70

Certifications

- Agentic AI Certified Foundations Associate — **Oracle**
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — **Oracle**
- Model Context Protocol: Advanced Topics — **Anthropic Academy**
- Complete Data Science, Machine Learning, Deep Learning, NLP Bootcamp — **Udemy**